5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*

SGSMA 2026 Smart Grid Synchronized Measurements and Analytics

Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

Jorge L. Mayorga Taborda, Yessica Africano Rodriguez, Fernando Jimenez

Universidad de los Andes | SGSMA 2026

Santiago de Chile · June 1–4, 2026

Date	Venue
June 1-4, 2026	Santiago, Chile

Motivation

Why frequency estimation in low-inertia IBR grids is hard

IBR disturbances expose a measurement gap

Event	Observed behavior	Monitoring limitation	Benchmark implication
Blue Cut Fire, CA 2016	Transmission faults caused nearly 1,200 MW of solar PV output loss; NERC linked part of the behavior to near-instantaneous frequency or voltage response.	The critical interval was a distorted, fault-clearing transient on a subsecond timescale.	Test whether estimators confuse waveform distortion with real frequency collapse.
Canyon 2 Fire, CA 2017	Two cleared faults produced 682 MW and 937 MW solar PV reductions.	NERC noted 4 s SCADA could not distinguish momentary cessation from tripping in some cases.	Report latency, settling, invalid samples, event-window behavior, and aggregate error.
Odessa events, TX 2021–2022	More than 1 GW-scale solar reductions followed normally cleared faults far from affected plants.	NERC found controls, protection, and model-fidelity gaps, including effects not captured well by positive-sequence models.	Stress tests must include control-like discontinuities, phase jumps, and model-mismatch regimes.
California BESS, 2022	Normally cleared faults caused abnormal BESS active-power reductions and inverter trips.	Causes included ac overcurrent/overvoltage, dc bus issues, unbalanced currents, and data-quality limits.	Evaluate frequency estimators under unbalance and non-sinusoidal contamination.

Source: NERC disturbance reports; NERC IBR disturbance monitoring white paper

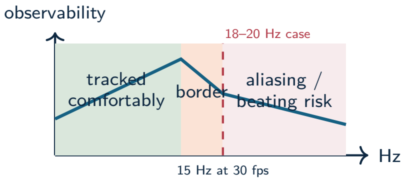
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i 2021 event exhibited 18–20 Hz oscillations after a generator trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

Source: Kaua'i 18–20 Hz IBR oscillation analysis; NERC IBR monitoring white paper; IEC/IEEE 60255-118-1

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Technical contributions

Not claimed here

- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- OpenFreqBench: a reproducible stress-test benchmark and platform for relay-class estimators;
- a metric-complete comparison protocol: RMSE, peak error, RoCoF error, T_{trip}, settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes;
- an adapted RA-EKF as a tested estimator, with explicit $\dot{\omega}$ state and event gating.

The benchmark explains why estimator rankings change across objectives, regimes, and compute budgets.

Experiment scale

18
estimators

4
families

5
scenarios

grid
search

Central finding

No single estimator wins across objectives: the ranking changes with scenario, metric, and compute budget. Protection-class frequency measurement needs tests beyond compliance-style cases.

Dataset: 18 estimators (incl. legacy baselines) across 33 scenarios, $n = 5$ Monte Carlo per scenario, with standard and IBR-specific metrics, plus ATLAS severity sweeps.

Estimator selection logic

Event regime
what stress dominates?

Ramp / RoCoF
phase jump
harmonics
multi-event

Estimator selection logic

Event regime
what stress dominates?

→

Objective metric
what failure matters?

Ramp / RoCoF
phase jump
harmonics
multi-event

RMSE
RoCoF error
T_{trip}
settling

Estimator selection logic

Event regime — what stress dominates? → **Objective metric** — what failure matters? → **Compute envelope** — what can run online?

Event regime	Objective metric	Compute envelope
Ramp / RoCoF	RMSE	MCU/DSP
phase jump	RoCoF error	FPGA
harmonics	T_{trip}	CPU/GPU
multi-event	settling	latency budget

Estimator selection logic

Event regime — what stress dominates? → **Objective metric** — what failure matters? → **Compute envelope** — what can run online? → **Evidence level** — replicated or pilot?

Event regime	Objective metric	Compute envelope	Evidence level
Ramp / RoCoF phase jump harmonics multi-event	RMSE RoCoF error T_{trip} settling	MCU/DSP FPGA CPU/GPU latency budget	full sweep (n=5) ATLAS severity (n=30 paper-grade) validation gate

Estimator selection logic

Event regime
what stress dominates?

Objective metric
what failure matters?

Compute envelope
what can run online?

Evidence level
replicated or pilot?

Ramp / RoCoF
phase jump
harmonics
multi-event

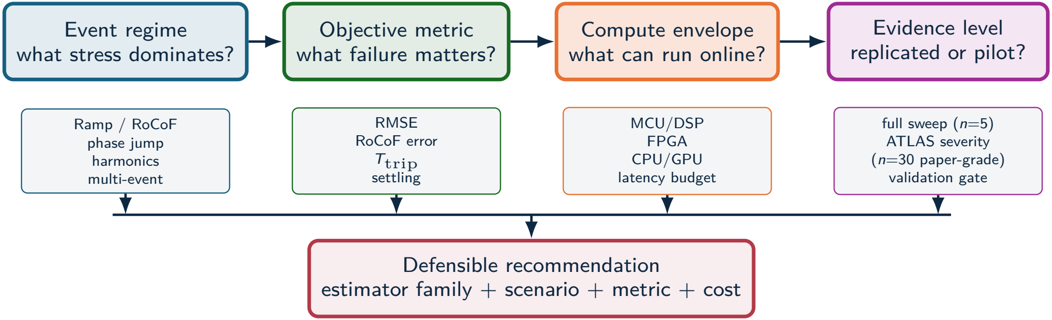
RMSE
RoCoF error
T_{trip}
settling

MCU/DSP
FPGA
CPU/GPU
latency budget

full sweep (n=5)
ATLAS severity
(n=30 paper-grade)
validation gate

Defensible recommendation
estimator family + scenario + metric + cost

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Benchmark & platform

What is missing in prior work — and OpenFreqBench

Benchmark literature gap

The field is mature in standards, metrology, open platforms, real datasets, and single-method studies. The missing layer is a reproducible, estimator-level benchmark that compares many algorithms under identical synthetic truth and deployment metrics.

Prior benchmark type	Strength	Gap for estimator-family comparison
IEC/IEEE PMU compliance	Defines synchrophasor, frequency, RoCoF, static/dynamic compliance tests	Leaves algorithm choice, family comparison, and CPU/latency outside the estimator report.
NIST PMU assessment/calibration	Provides metrological evaluation of PMU devices against standard requirements	Focuses on instruments and compliance behavior rather than a large matrix of open estimator implementations under controlled tuning policy.
OpenPMU / openPDC ecosystem	Enables open PMU construction and synchrophasor stream processing	Infrastructure-oriented, with no controlled estimator matrix using matched seeds and stress sweeps.
Open real PMU datasets	Supply realistic events for detection and application development	Truth is partially latent; hard to isolate causal stress factors or compute estimator error against known $f_{\text{true}}[k]$.
Single-study algorithm comparisons	Give detailed evidence for selected methods or scenarios	Usually limited estimator sets, limited artifact contract, and incomplete deployability metrics.

Compliance tests define requirements, operational data define realism, and controlled synthetic truth isolates estimator behavior under reproducible stress.

Source: IEC/IEEE 60255-118-1:2018; NISTIR 8106; OpenPMU; openPDC; FNET/GridEye; LBNL Open micro-PMU; PNNL open PMU event library

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

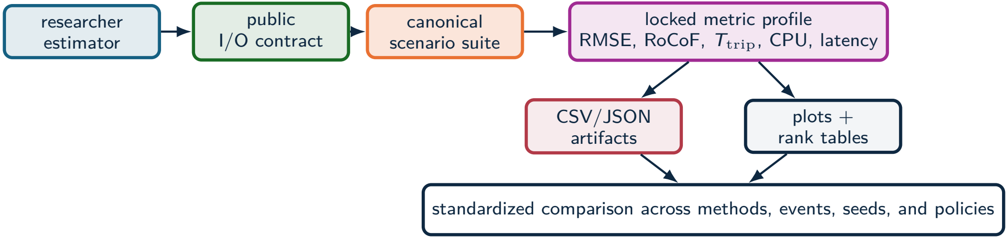
Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

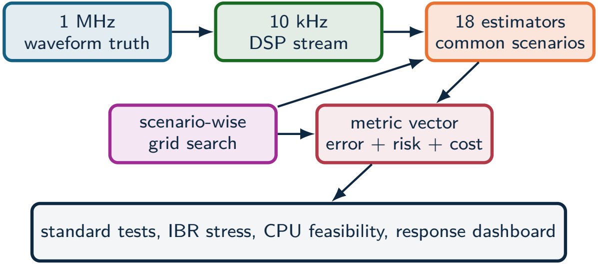
Benchmark reports, metric tables, plots, manifests, and readiness gates.

> OpenFreqBench turns a new estimator into a repeatable benchmark entry instead of a one-off comparison.

Experimental method

Same truth, same metrics, fair tuning

Experimental method



Measurement setting

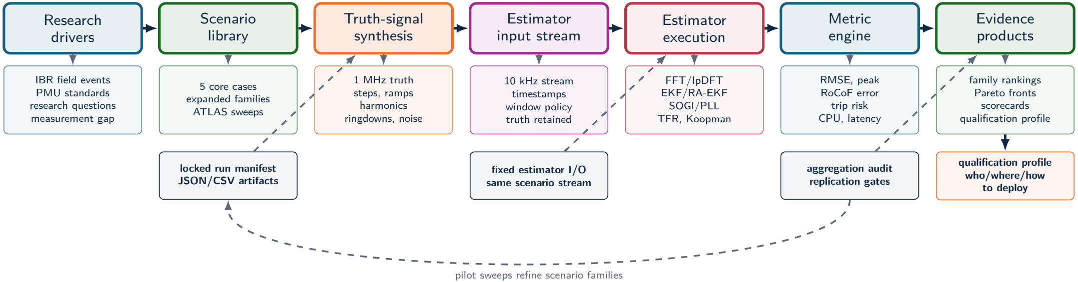
Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

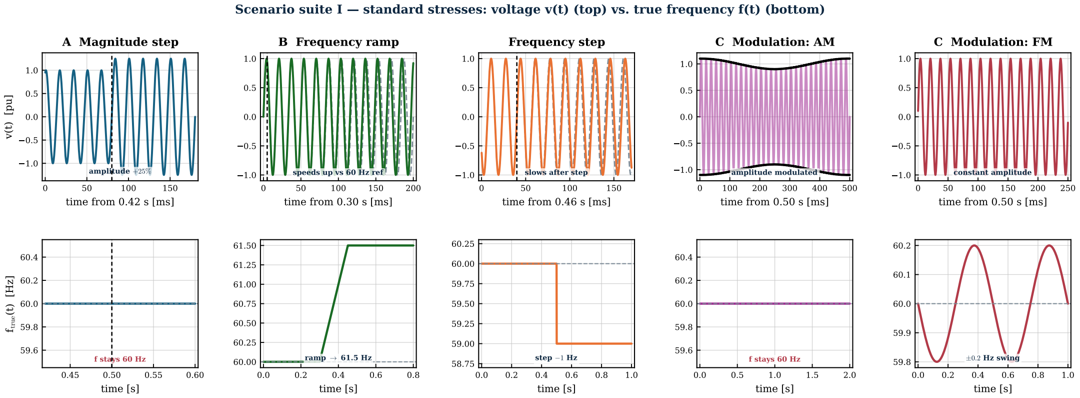
Source: Methodology and test framework

Full benchmark workflow

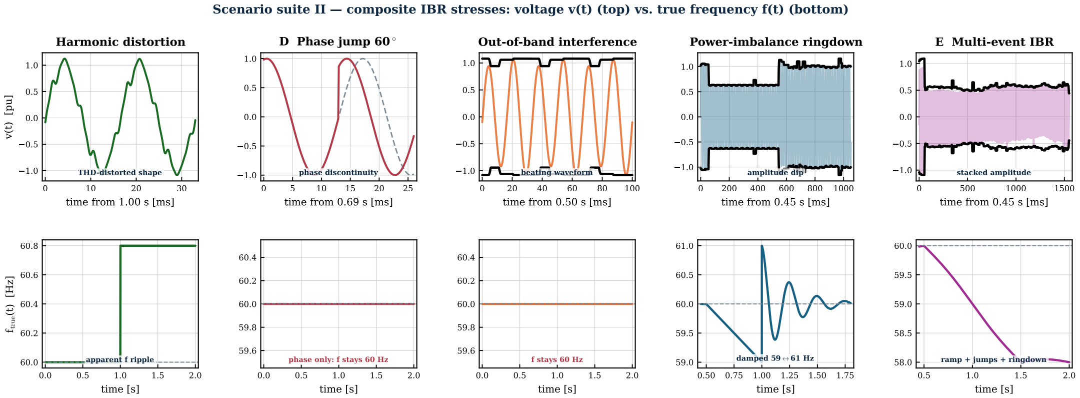


The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

Scenario suite I — standard stresses



Scenario suite II — composite IBR stresses



Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Legacy baselines

TKEO and ZCD are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\mathrm{true}}[k]$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\max} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\mathrm{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\ \mathrm{Hz}\}$$

$$t_s = \min\{t : \ |e[\tau]| \leq \epsilon,\ \forall \tau \geq t\}$$

$$C_{\mathrm{CPU}} = \mathrm{mean\ time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Core evidence

From standard cases to composite IBR stress

Standard scenario results

Method	Step +10% RMSE [Hz]	Ramp RMSE [Hz]	Modulation 2 Hz RMSE [Hz]
IpDFT	**0.000**	**14.4**	0.114
TFT	0.001	0.019	0.046
Koopman	0.003	0.017	0.044
PLL	0.033	**1.47**	**0.227**
SOGI-FLL	0.005	0.013	0.040
EKF	**0.000**	**37.1**	0.064
UKF	**0.000**	0.017	0.055
RA-EKF	0.000	**0.006**	0.037
LKF2	0.138	0.015	**0.028**

Reading (mean over 5 seeds): RA-EKF, SOGI-FLL, Koopman and TFT stay $\leq$ 0.05 Hz on all three, but **EKF and IpDFT intermittently diverge on the ramp** (37 and 14 Hz) and PLL degrades on ramp/modulation. Robustness, not point accuracy, separates the families.

Source: `full_mc_benchmark`, $n = 5$

Scenario D: phase jump separates filters from windowed methods

State-space + loops recover cleanly

A clean $+60^\circ$ phase jump (NERC profile) is absorbed with near-zero error and **zero trip exposure**:

- EKF, UKF, RA-EKF: RMSE ≈ 0 Hz, $T_{\mathrm{trip}} = 0$.
- PLL 0.021 Hz, SOGI-FLL 0.067 Hz, $T_{\mathrm{trip}} = 0$.

Windowed / spectral methods trip

The trip exposure here lives in the window/spectral family:

- IpDFT: 1.56 Hz RMSE, $T_{\mathrm{trip}} = 1.21$ s.
- RLS 2.28 Hz / 1.30 s; Prony 2.39 Hz / 1.35 s.
- TFT 0.43 Hz / 0.09 s; Koopman 0.39 Hz / 0.11 s.

Benchmark reading: an isolated phase jump does *not* break the state-space filters — EKF already recovers with zero trip exposure. RA-EKF's real advantage appears on ramps and multi-event stress (next), not on clean phase jumps. Source: `full_mc_benchmark`, $n = 5$: `NERC_Phase_Jump_60`

Composite IBR sequence: why it is hardest

Scenario E stress components

- Harmonic-rich steady state: 5th at 5%, 7th at 3%, low interharmonics.
- $+40^\circ$ phase jump.
- -6 Hz/s RoCoF segment.
- $+80^\circ$ phase jump with ringdown.
- Sparse impulsive noise.

Standard-test gap

- The estimator sees a nonstationary waveform with stacked disturbances.
- Spectral leakage, loop re-lock, state-model mismatch, and recovery delay occur together.
- A method can be good on RMSE and still undesirable on trip exposure or CPU.

Once composite stress is shown to matter, ATLAS sweeps each stress component systematically instead of treating the multi-event case as a single aggregate disturbance.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
UKF	**1.09**	**0.78**	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	**0.78**	11
SOGI-FLL	1.14	0.84	**9**
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	**15 241**
ZCD	**426**	1.55	7

No accurate winner

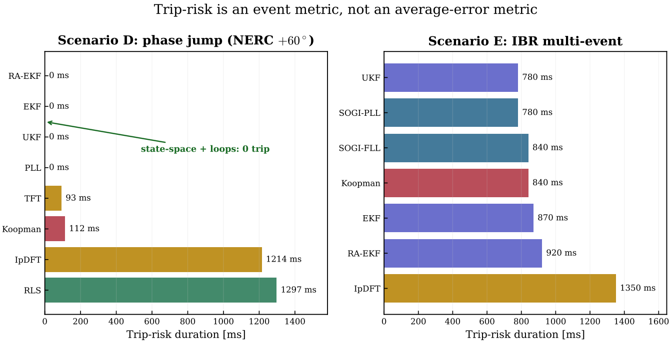
Under the stacked sequence every usable method collapses to $\approx$ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000$\times$ costlier; RA-EKF sits mid-pack but never diverges.

Source: `full_mc_benchmark`, $n = 5$: `IBR_Multi_Event`

Trip-risk evidence: core stress cases

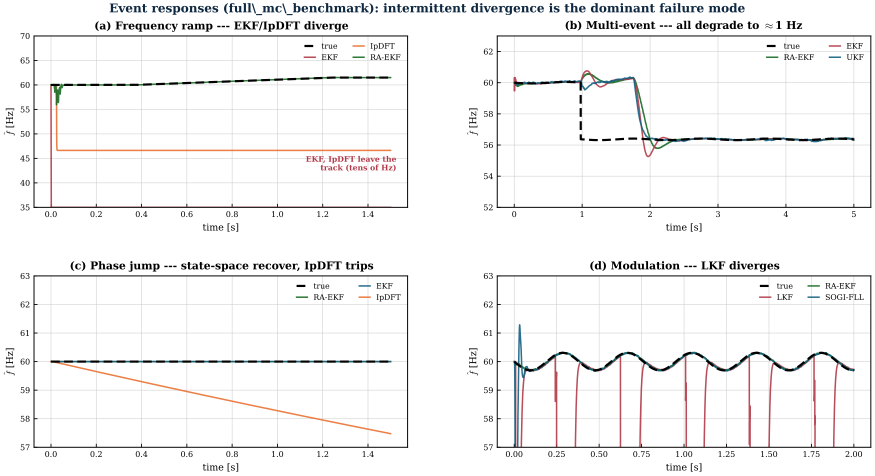


Result reading

- Scenario D: state-space filters and PLL keep $T_{\text{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: *every* method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

Event responses



Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step $\approx$ 0; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all $\approx$ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000$\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 μs/sample; Koopman $\sim$15,000	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample	Family	Target
ZCD	7 μs	Loop	MCU
SOGI-FLL	9 μs	Loop	MCU
PLL	9 μs	Loop	MCU
EKF	11 μs	Model	MCU/DSP
UKF	13 μs	Model	DSP
RA-EKF	**15 μs**	**Model**	**DSP**
IpDFT	17 μs	Window	DSP/FPGA
ESPRIT	$\approx$1,400 μs	Window	CPU
Koopman	$\approx$15,000 μs	Data-driven	CPU/GPU

Source: `full_mc_benchmark`, $n = 5$: `mean m13_cpu_time_us`

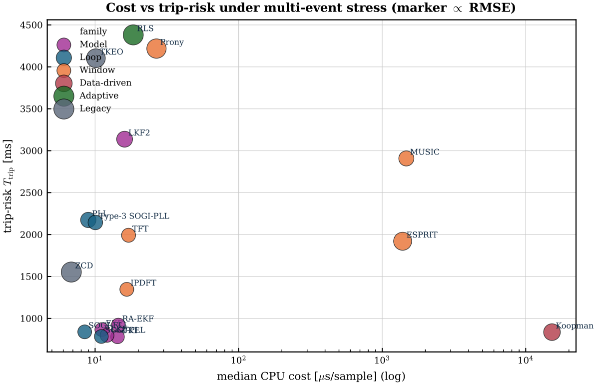
Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

Key comparison

The state-space and loop families all run at $\sim$10–17 μs/sample (embedded-feasible); spectral (ESPRIT, MUSIC) and data-driven (Koopman) methods are 100–1000$\times$ costlier and data-dependent.

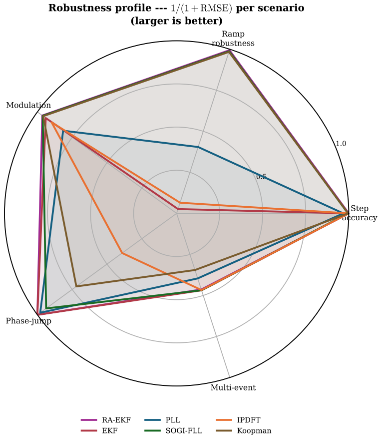
Deployment limits



Dashboard reading

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- State-space + loop methods cluster at low cost ($\leq 17\,\mu$s) and low trip.
- Spectral/data-driven methods (MUSIC, ESPRIT, Koopman) are 100–1000$\times$ costlier; RLS/Prony pair high trip with low accuracy.
- RA-EKF sits in the low-cost, never-diverging region.

Balance profile



Radar profile

- Each axis is one scenario; larger = more robust ($1/(1 + \mathrm{RMSE})$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/T_{trip}/settling threshold | **F*** borderline.

No single column predicts the others. EKF and IpDFT pass step/modulation yet **intermittently fail the ramp** (divergence); windowed/spectral methods fail the phase jump; and *no* method survives the multi-event sequence. Standard tests alone leave IBR readiness untested.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode. On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\text{trip}} = 0$.

Metric disagreement

RMSE, trip-risk, and CPU select different winners. Under multi-event stress UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved. The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Deployability

Feasibility changes the accuracy ranking. RA-EKF runs at $\sim$15 μs/sample; the spectral and data-driven methods are 100–1000× costlier and sit outside relay-class timing.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Expanded & ATLAS

Does the pattern scale across families and severity?

Expanded stress analysis

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy, paper-grade $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Expanded benchmark: family winners

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	RA-EKF	11 mHz	0 s (state-space)
Magnitude steps	7	UKF	<1 mHz	0 s (state-space)
Modulation	3	LKF2	15 mHz	0 s (state-space)
Phase jumps	3	UKF	<1 mHz	0 s (state-space)
Frequency step	1	RA-EKF	52 mHz	RA-EKF, 3 ms
OOB interference	1	UKF	<1 mHz	0 s (state-space)
IBR harmonics	3	SOGI-FLL	61 mHz	SOGI-PLL, 10 ms
IBR ringdown	4	LKF2	202 mHz	ESPRIT, 87 ms
IBR multi-event	1	UKF	1.09 Hz	UKF, 780 ms

The full Monte-Carlo sweep confirms the central pattern: the RMSE leader rotates (RA-EKF, UKF, LKF2, SOGI-FLL) by disturbance family, and error grows from sub-mHz on clean cases to ∼1 Hz under multi-event stress.

ATLAS severity analysis

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

Core cases → ATLAS severity curves → sign asymmetry → fixed vs oracle policy → validation gate

Core cases	ATLAS severity curves	sign asymmetry	fixed vs oracle policy	validation gate
A–E failure mode	When does error start to grow?	Are positive and negative events symmetric?	Does tuning policy change the rank?	Which results replicate?

ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: `docs/ATLAS_PIPELINE.md`; `docs/ATLAS_METHOD_AUDIT.md`

Reading the evidence: three levels

Diagnostic ($n < 30$)
mechanism + debugging only

↓

Paper-grade ($n \geq 30$)
fixed policy, both signs

↓

Journal-grade ($n \geq 100$)
cost reps, archived hashes

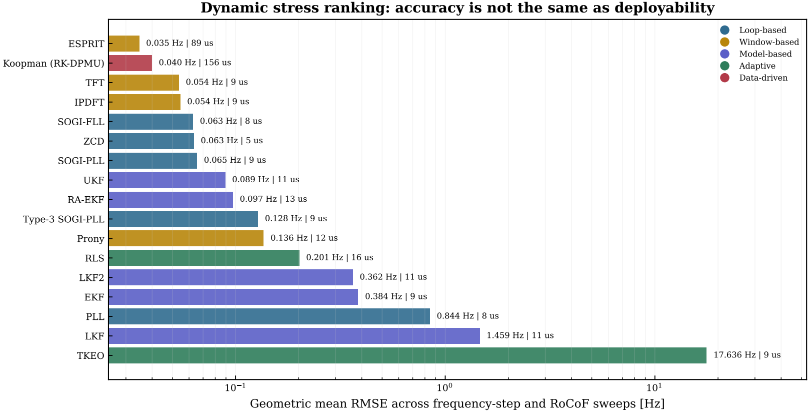
Where this talk sits

- Core 5-scenario results: tuned comparison evidence ($n = 5$).
- ATLAS severity sweeps: **paper-grade** — $n = 30$, fixed policy, both signs, **9/9** required sweeps (full-ATLAS readiness met).
- Journal-grade ($n \geq 100$, archived hashes) is the remaining step.

The ATLAS severity results are now paper-grade; the journal-grade campaign ($n \geq 100$) and the subspace/ML slow-estimator sweep are the remaining steps.

Dynamic accuracy ranking

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



When the textbook filter diverges

DIAGNOSTIC PILOT: small-n Monte Carlo, per-scenario tuned — paper-grade $n \geq 30$ run pending

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

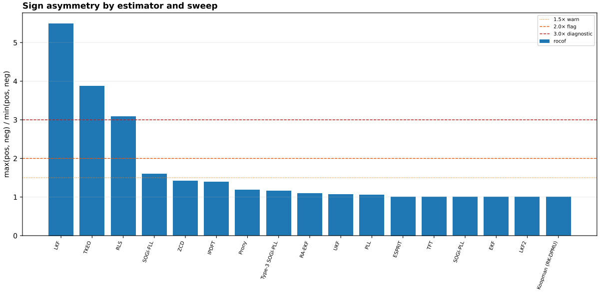
Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

UKF shows the same intermittent divergence on the 5 and 15 Hz/s ramps; the robustified state-space variant stays stable across the whole ramp family.

RoCoF has a sign: down-ramps are harder

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Source: `atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv` ($n = 30$, both signs)

Directional bias ($RMSE_{-}/RMSE_{+}$ at ± 50 Hz/s)

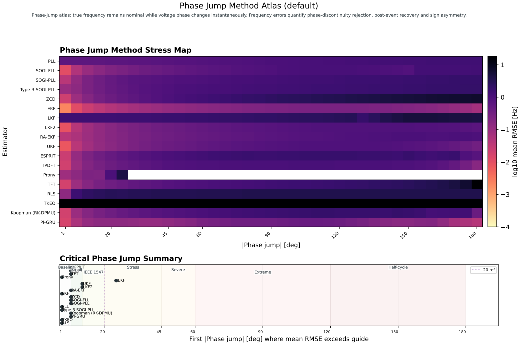
- LKF **5.5**× worse on −RoCoF
- TKEO **3.9**×, RLS **3.1**×
- SOGI-FLL 1.6×
- EKF, UKF, RA-EKF ≈ 1.0 (symmetric)

Why it matters

A +RoCoF-only test can certify a method that fails on frequency *decline* — the protection-critical direction. Both signs must be swept.

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only **0.25 Hz @ 115°**, back to 0.07 Hz @ 180°.

The TFT cliff

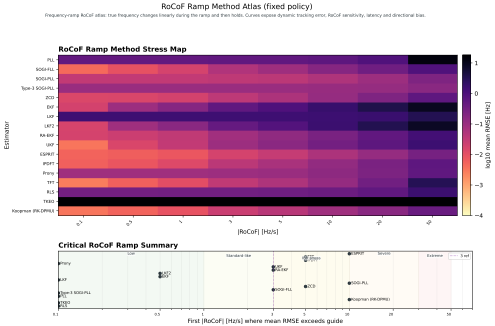
One exception: TFT rises gently, then jumps **2.5 → 21.9 Hz** between 150° and 180° — a phase-reversal singularity.

"Bigger jump = worse" is wrong for this family. Severity sweeps, not single test points, expose where each estimator actually breaks.

No universal winner: champions rotate by regime

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	4.1 mHz
Standard (1–3)	ESPRIT	7.6 mHz
IBR stress (3–10)	ESPRIT	22.3 mHz
Severe (10–30)	SOGI-PLL	70.4 mHz



Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.

The deliverable is a selection map — event class × metric × compute budget — not a single ranking.

The ZCD paradox: best and worst, same estimator

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Low-stress champion

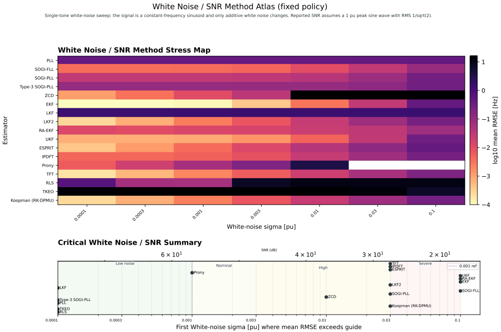
Up to ∼15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.

Broadband-noise cliff

Same method: 34 mHz at $\sigma = 0.003$ pu → **3112 Hz** at $\sigma = 0.1$ pu. EKF stays ≤ 0.43 Hz across the same sweep.

Harmonic cliff

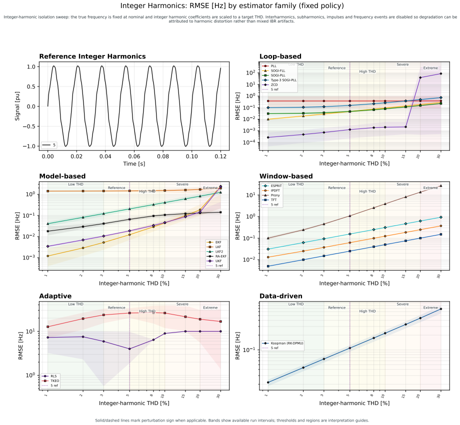
Same method again: 2.2 mHz at 15% THD → **39 Hz** at 20% → **86 Hz** at 30%.



Aggregate rankings hide stress-specific collapse: the same estimator is the best and the worst depending on the disturbance. The benchmark reports per-stress behavior, not one score.

Harmonic degradation differs by estimator family

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

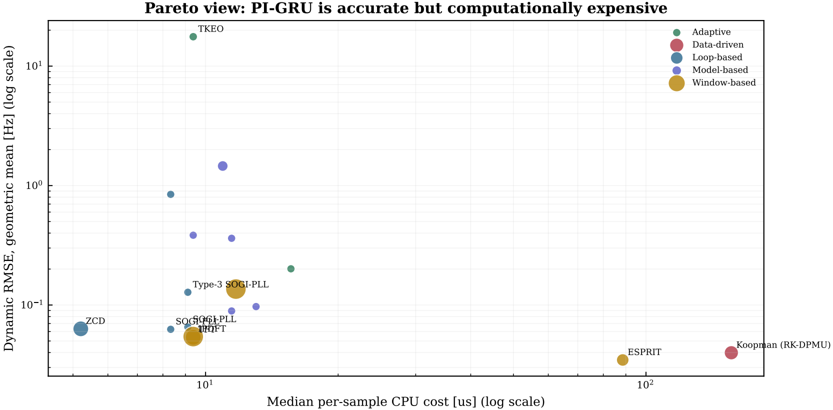


Reading

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above ∼15%; window methods (TFT) degrade gently and lead at extreme THD.

CPU–accuracy Pareto analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Findings & deployment

What to deploy — and what comes next

Three findings

1. **Latency and stress trade-off**: IpDFT/TFT provide strong steady-state accuracy but carry 3–5 cycle structural delay; loop methods are fast but can suffer elevated trip exposure.

Three findings

1. **Latency and stress trade-off**: IpDFT/TFT provide strong steady-state accuracy but carry 3–5 cycle structural delay; loop methods are fast but can suffer elevated trip exposure.
2. **Robustness beats point accuracy**: EKF, UKF and IpDFT intermittently diverge on ramps and multi-event stress (tens of Hz on individual seeds), while RA-EKF's explicit $\dot{\omega}$ state keeps it stable across every seed.

Three findings

1. **Latency and stress trade-off**: IpDFT/TFT provide strong steady-state accuracy but carry 3–5 cycle structural delay; loop methods are fast but can suffer elevated trip exposure.
2. **Robustness beats point accuracy**: EKF, UKF and IpDFT intermittently diverge on ramps and multi-event stress (tens of Hz on individual seeds), while RA-EKF's explicit $\dot{\omega}$ state keeps it stable across every seed.
3. **Compliance tests are insufficient**: estimators that pass standard-style Step/Ramp/Mod cases can fail under composite islanding and multi-event IBR stress.

Three findings

1. **Latency and stress trade-off**: IpDFT/TFT provide strong steady-state accuracy but carry 3–5 cycle structural delay; loop methods are fast but can suffer elevated trip exposure.
2. **Robustness beats point accuracy**: EKF, UKF and IpDFT intermittently diverge on ramps and multi-event stress (tens of Hz on individual seeds), while RA-EKF's explicit $\dot{\omega}$ state keeps it stable across every seed.
3. **Compliance tests are insufficient**: estimators that pass standard-style Step/Ramp/Mod cases can fail under composite islanding and multi-event IBR stress.

Estimator qualification includes transient recovery, trip exposure, compute feasibility, and standard-case RMSE.

Validity limits and next steps

Validation limits

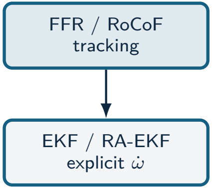
- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

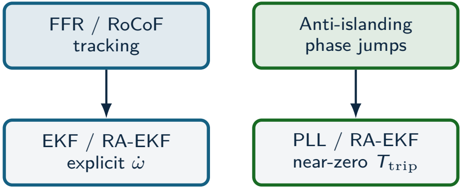
- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

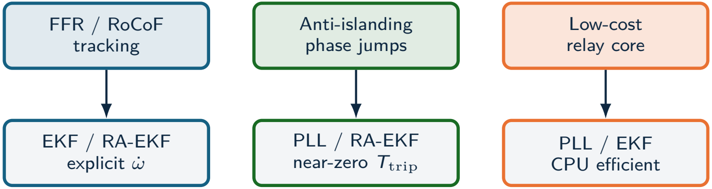
Recommendation map



Recommendation map



Recommendation map



Recommendation map

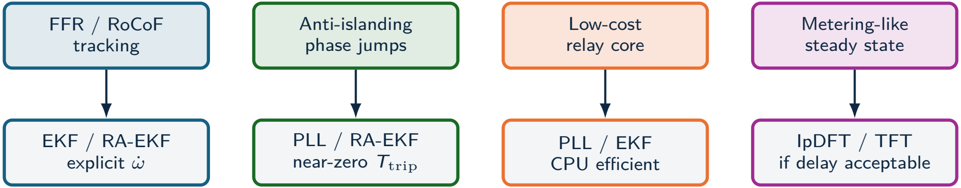
FFR / RoCoF tracking → EKF / RA-EKF explicit $\dot{\omega}$

Anti-islanding phase jumps → PLL / RA-EKF near-zero T_{trip}

Low-cost relay core → PLL / EKF CPU efficient

Metering-like steady state → IpDFT / TFT if delay acceptable

Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

Qualification profile

Core benchmark → Conditional estimator ranking → Metric-weighted event score → Dynamic estimator benchmark profile

Known truth signals → Estimator I/O contract → Locked metric registry → Machine-readable evidence

Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \, \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

- $\tilde{E}_s$: normalized dynamic RMSE.
- $\tilde{P}_s$: normalized peak deviation.
- $\tilde{T}_s$: trip-risk duration.
- $\tilde{S}_s$: settling and recovery penalty.
- $\tilde{C}_s, \tilde{L}_s$: compute and latency penalties.

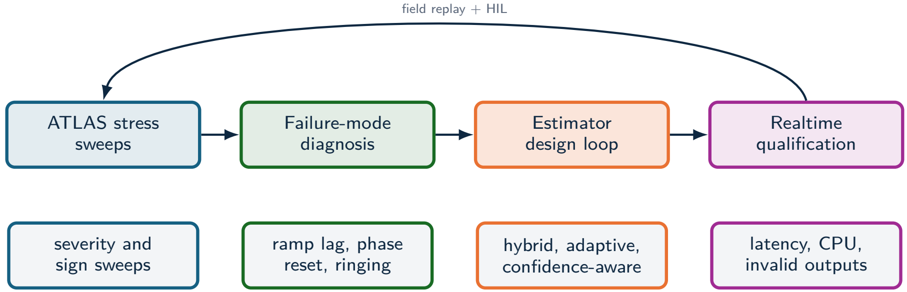
Composite metric

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score states the deployment objective explicitly.

Profile-specific weights

Protection-adjacent use can raise γ, δ, λ; forensic reconstruction can raise α, β; embedded monitoring can raise η.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Ongoing and future work

Full-microgrid fault validation [in progress]

Integrate Simulink simulation outputs (CSV) from the Chamorro *et al.* complete-microgrid fault model, replaying realistic fault waveforms through the same estimator pool.

C++ acceleration

Port the heavy estimators and 1 MHz signal generation to native C++ modules to cut benchmark runtime and enable journal-grade ($n \geq 100$) Monte Carlo at scale.

Modern and ML-based estimators

Extend the pool with recent literature methods and machine-learning estimators, benchmarked under the identical metric contract.

ANDES co-simulation

Couple OpenFreqBench with the ANDES power-system dynamics engine to measure estimator performance under more realistic fault transients and control interactions.

These extensions move OpenFreqBench from synthetic stress tests toward field-realistic, validated estimator qualification.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to the event class, metric target, and compute envelope; the next validation step evaluates each method against declared event regimes, latency limits, and compute envelopes.

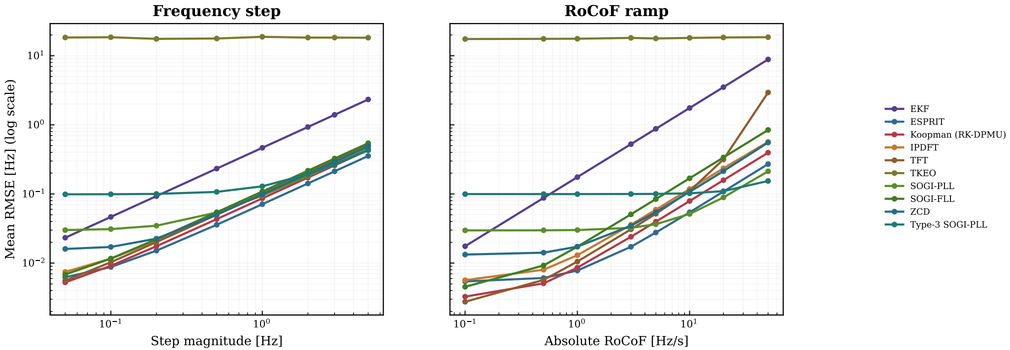
Backup

Metrics, extra stress maps, the journal-grade plan, references.

Sensitivity to disturbance severity

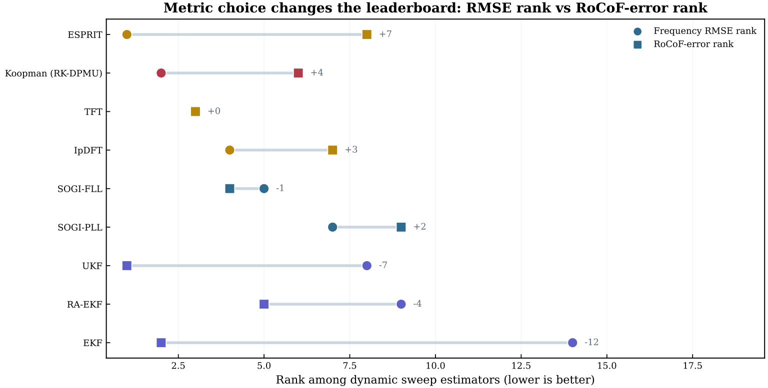
ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Severity response curves reveal different failure modes



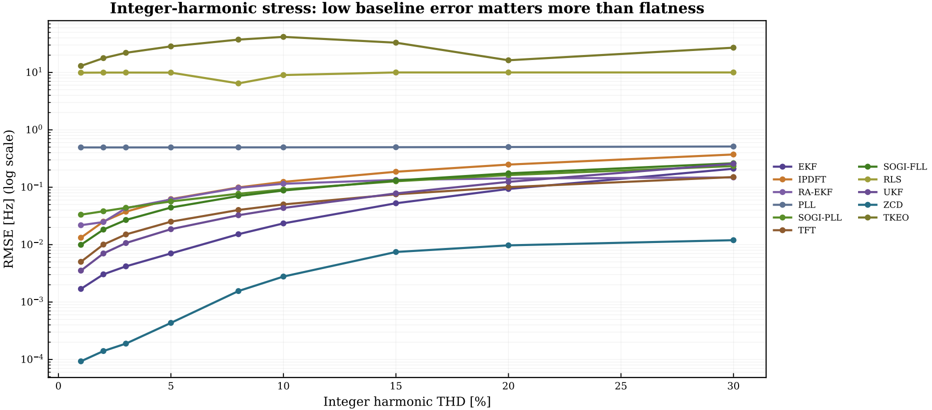
Rank shift when the metric changes

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Harmonic-stress response

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Backup

Backup: metric registry

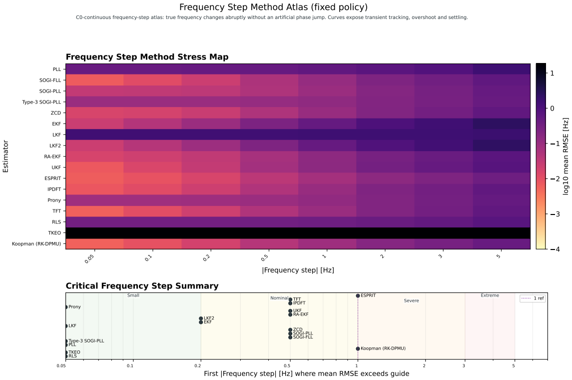
Metric	Definition
M1 RMSE	$\sqrt{\text{mean }e[k]^2}$ on steady-state-trimmed frequency error [Hz].
M3 peak	$\max_k \lvert e[k] \rvert$ [Hz].
M5 trip-risk	cumulative time with $\lvert e[k] \rvert > 0.5$ Hz (protection-deadband risk proxy) [s].
M6 max-contig trip	longest *contiguous* deadband violation [s].
M8 settling	time until $\lvert e \rvert$ stays ≤ 0.2 Hz [s].
M9 / M10 RFE	RoCoF error: robust max (99.5th pct) / RMS of $\lvert \widehat{\dot{f}} - \dot{f} \rvert$ [Hz/s].
M13 CPU	execution time per sample [μs] — software-cost proxy, not hardware latency.
M14 latency	declared structural delay (window / group delay) [ms].
M22 invalid rate	fraction of samples the estimator flags invalid / NaN.

Trip-risk is a protection-oriented risk proxy, not a relay trip command; CPU is a software-cost proxy. Differences below sampling/timing resolution are not interpreted.

Source: `src/analysis/metrics.py; docs/METHODS.md`

Backup: frequency-step stress map ($n = 30$, fixed policy)

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

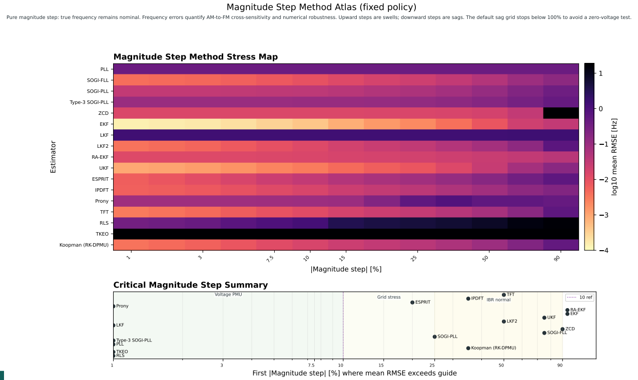


Reading

- **ESPRIT leads** almost every step; Koopman wins only the tiny ± 0.05 Hz steps.
- RMSE grows $\sim$linearly with step size ($\approx$0.07 Hz per 1 Hz).
- **TKEO flat-broken** ($\approx$15–22 Hz).
- **Sign-symmetric**: 16/17 within 1.5$\times$ — unlike magnitude steps and RoCoF.

Backup: magnitude-step stress map ($n = 30$, fixed policy)

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

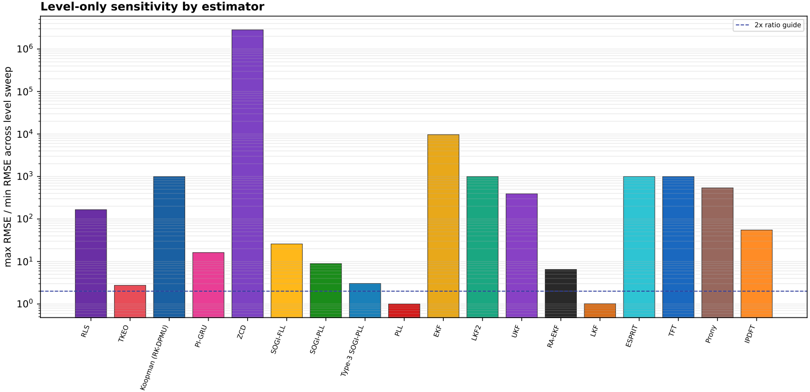


Reading

- **EKF leads** every step size (0.17–7.8 mHz).
- **TKEO is flat-broken** ($\approx$18 Hz at all levels).
- **ZCD fails on deep sags**: 374 Hz at -90% vs 0.009 Hz at $+90\%$.
- Voltage *sags* are $\sim 2\times$ harder than swells of equal size.

Backup: noise robustness ranking

DIAGNOSTIC PILOT: small-n Monte Carlo, per-scenario tuned — paper-grade $n \geq 30$ run pending



Backup: from paper-grade (done) to journal-grade

Completed — paper-grade

$n = 30$, fixed policy, both signs, **9/9** required sweeps; 17-estimator canonical set (PI-GRU experimental, separate sweep). Readiness gate: `status=paper_grade`, `scope=full_atlas`.

Next — journal-grade

```
-sweeps all -policy fixed_policy -n-runs 100
-n-cost-reps 3 -tune-trials 80 -output-subdir
atlas-full-v1
```

- Paper-grade $n = 30$ campaign: **done** (readiness gate passed).
- Journal-grade lifts to $n \geq 100$ with archived seeds, manifests, and hashes.
- Subspace/ML methods (MUSIC, ESPRIT, PI-GRU) run as a separate full-ATLAS sweep.
- Every table and figure regenerates from artifacts.

Paper-grade ($n = 30$, full ATLAS) is complete; the $n \geq 100$ journal-grade run and the subspace/ML sweep are the remaining steps.

Source: `docs/ATLAS_PIPELINE.md`; `docs/ATLAS_ROADMAP.md`

Backup: selected references

Standards & platforms

- IEEE/IEC 60255-118-1:2018 — synchrophasor, frequency, RoCoF compliance.
- NISTIR 8106 — PMU performance assessment.
- OpenPMU, openPDC, FNET/GridEye, LBNL micro-PMU, PNNL PMU event library.

IBR field events

- NERC Blue Cut Fire (2016), Canyon 2 (2017), Odessa (2021–22), CA BESS (2022).
- NERC IBR disturbance-monitoring white paper.
- Kaua'i 2021 18–20 Hz IBR oscillation analysis.

Field cases define the stress modes; no claim is made that a single estimator would have prevented a disturbance.